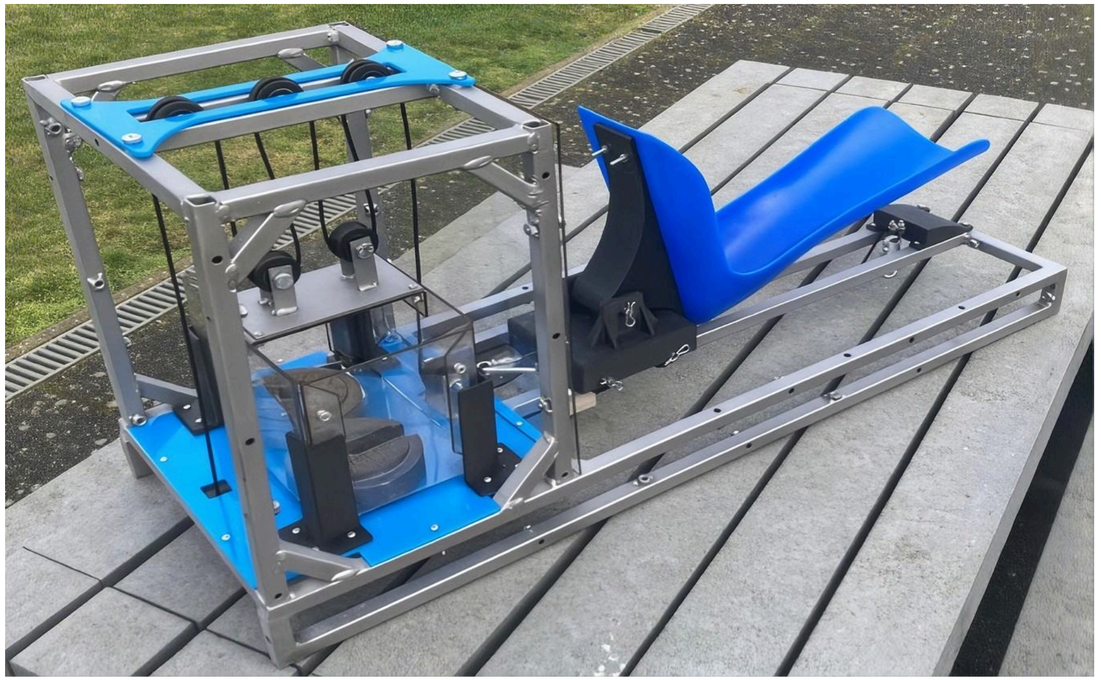
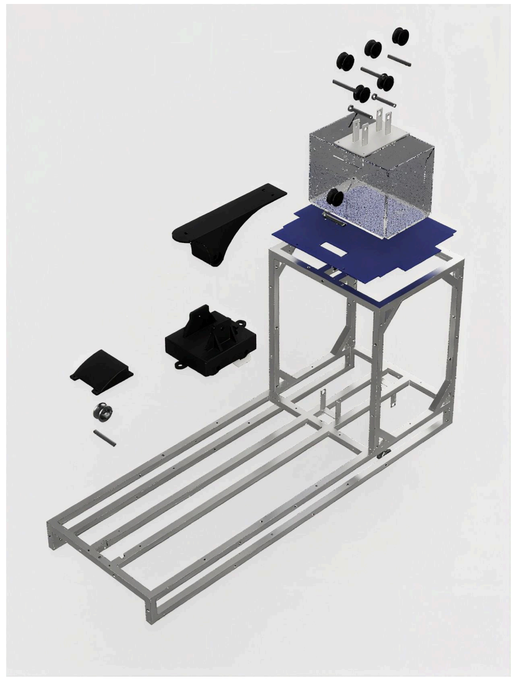
ICU Rehabilitation Device

A-Level DT, 2024

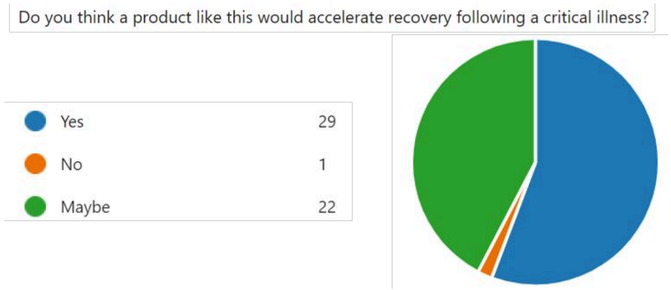
Designed and built for A-Level Design & Technology: a rehabilitation device for ICU patients facing muscular atrophy. The final prototype features a pulley resistance system, an adjustable weight box, bidirectional resistance, a freely rotating boot attachment and a removable patient-specific boot which helps prevent foot-drop. Development ran from early wood and cardboard proof-of-concept prototypes through CAD modelling to the final steel and acrylic build.











Developed with input from an occupational therapist, and validated through a survey of over 50 medical professionals across the ICU and St Bartholomew's Hospital, most of whom rated it likely to be useful and to accelerate recovery.

Luka Hinton

Eames Kneeling Chair

Personal, 2026

A rocking kneeling chair that combines the Varier Variable's rocking geometry (Peter Opsvik, 1979) with Eames-style moulded plywood shells, fusing classic lounge furniture with ergonomic work seating. The final design makes use of powder-coated cast aluminium, walnut-faced moulded ply, rubber shock mounts and leather upholstery. My own contributions are a tip-and-roll transport feature, and anthropometric checks against P5 and P95 users that exposed a foot clash and drove a frame redesign. Modelled, analysed and rendered in Fusion, with FEA under BS EN seating test loads.

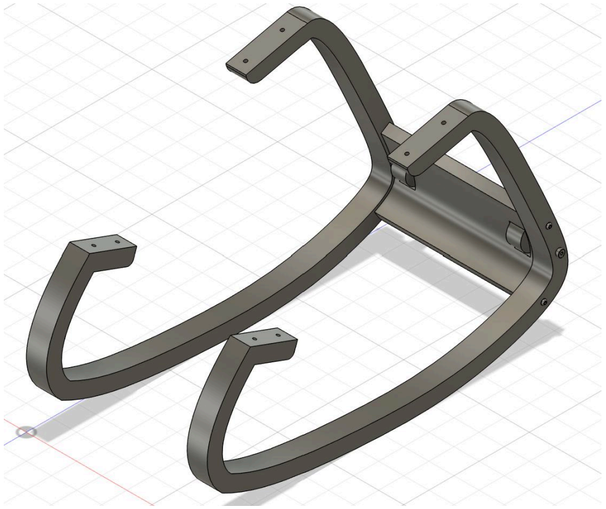


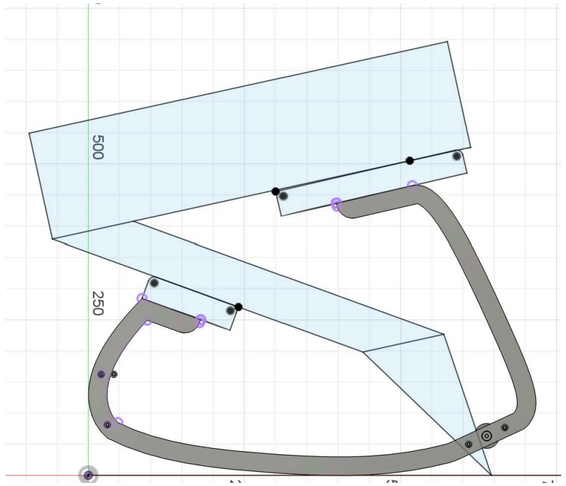




Eames Kneeling Chair

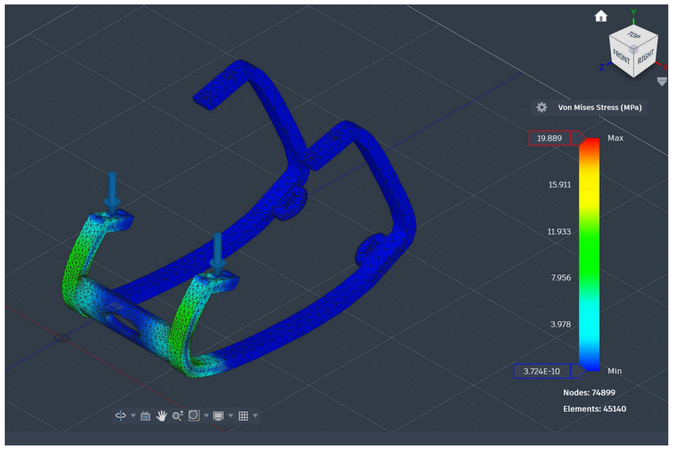
Personal, 2026

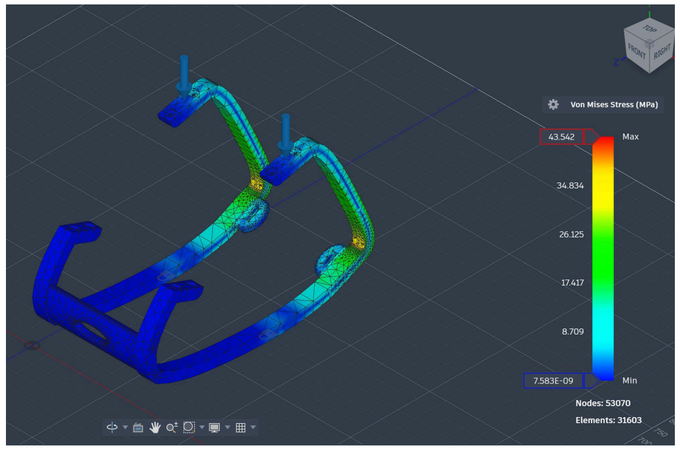






The original design led to a foot clash when testing with a P95 model, leading to a redesign of how the wheels were attached.



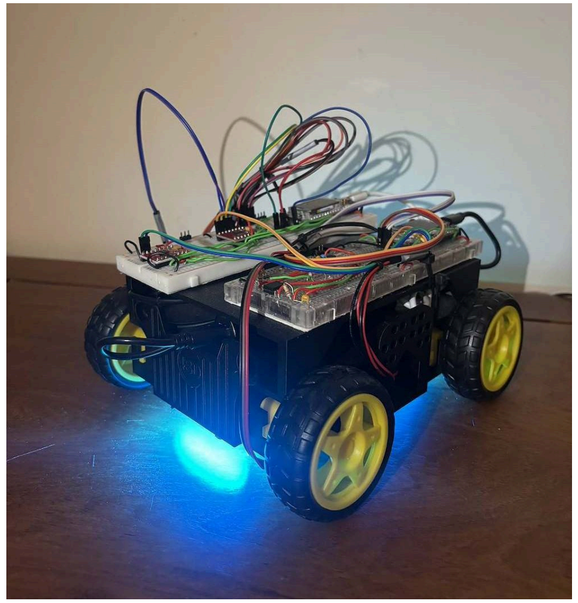


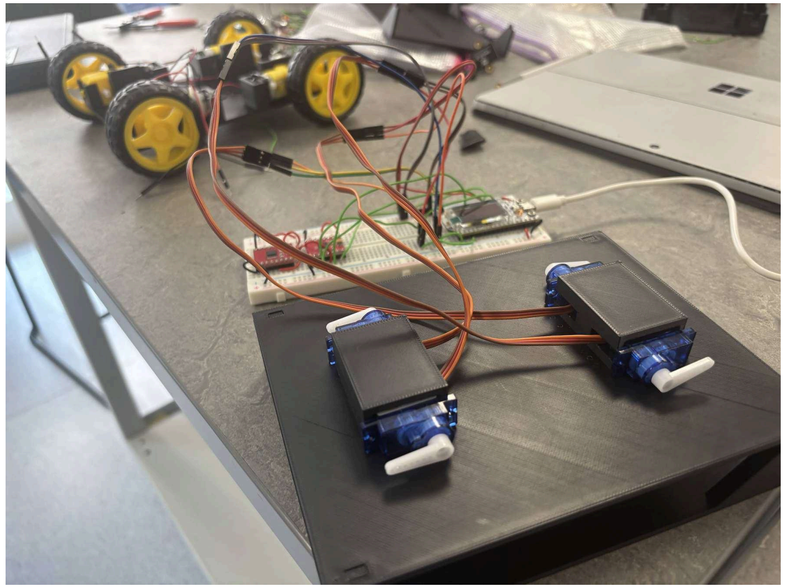


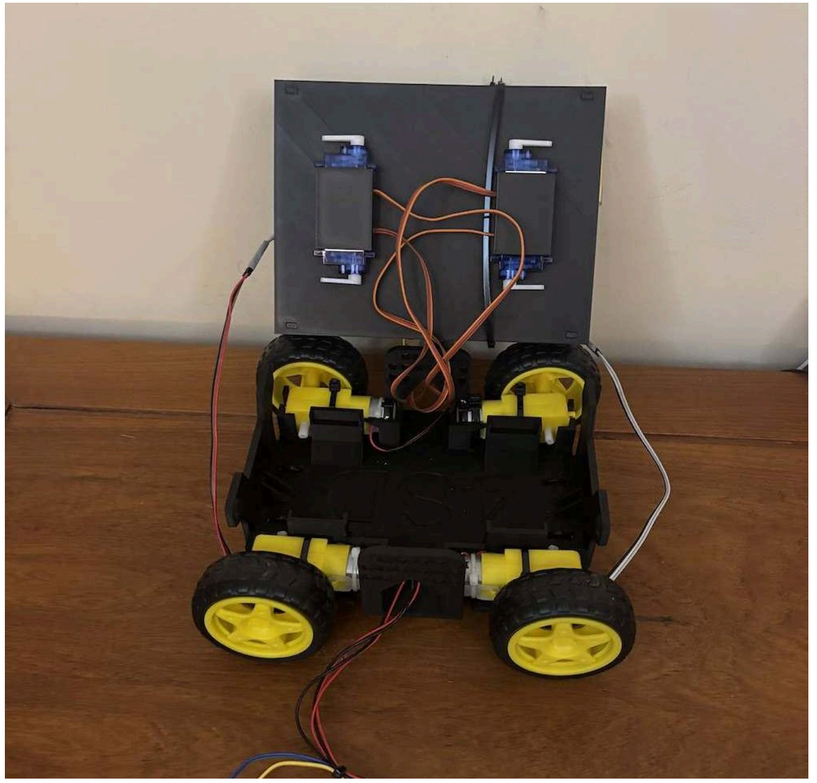
RC Car

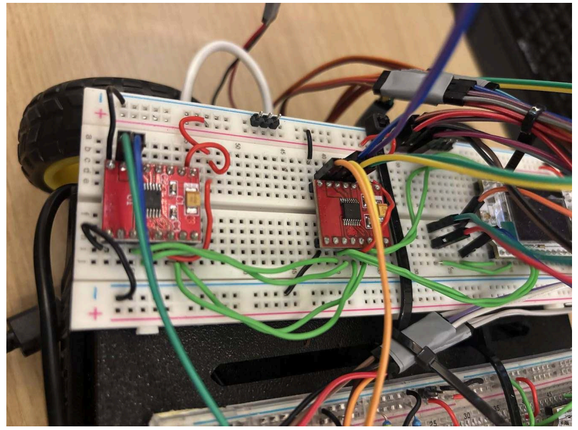
Electronics 1, 2026

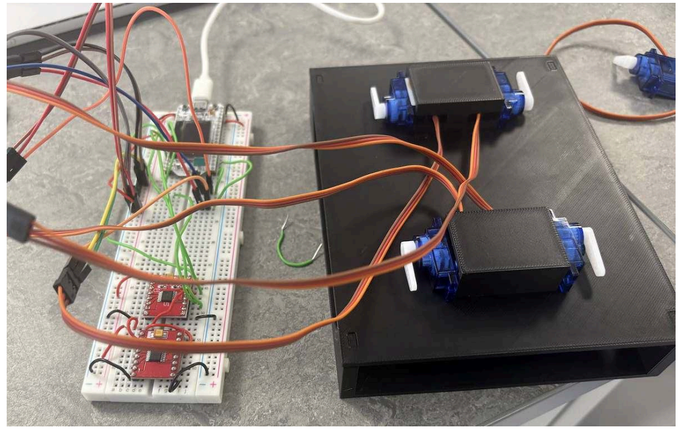
Built for my Electronics 1 module: an RC car driven over WiFi from a phone. Servo motors control ride height, giving a hydraulic-style 'bounce' similar to a Maybach, and underskirt LED lighting reacts live to nearby music, syncing to the beat.











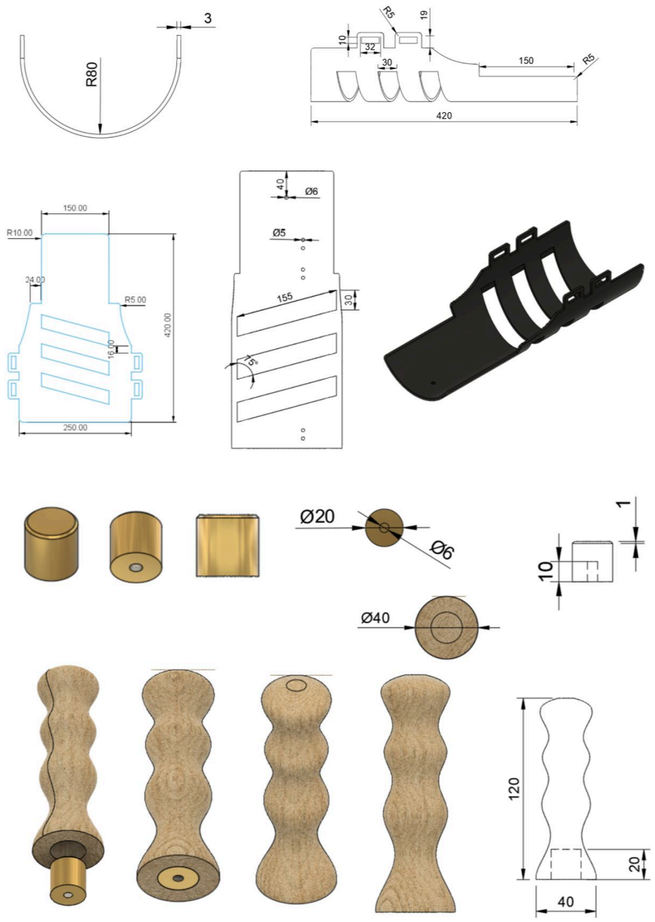
Luka Hinton

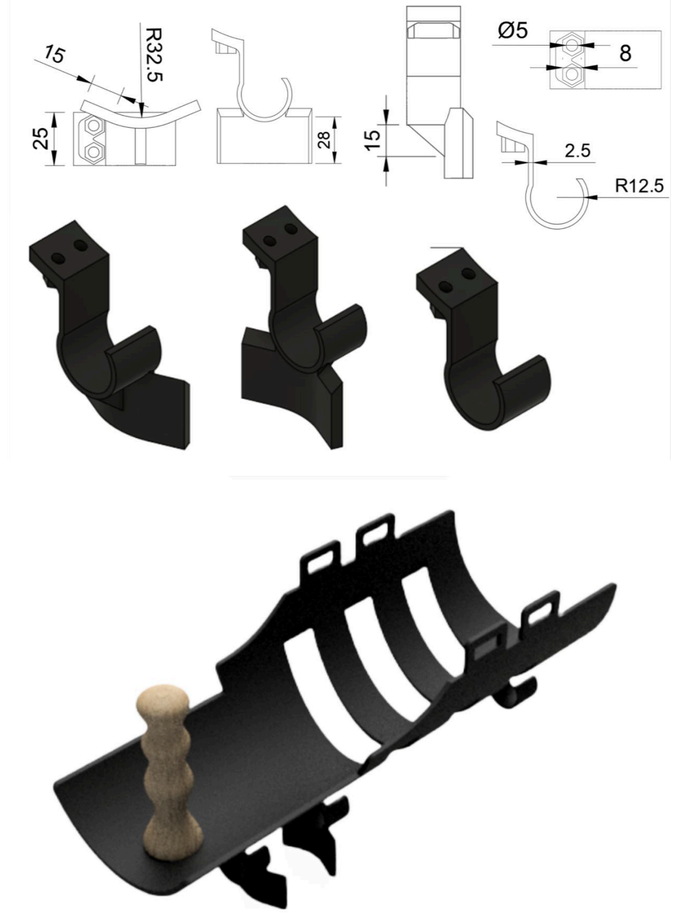
Fishing Arm Support

GCSE DT, 2022

Designed and built for GCSE Design & Technology: an arm support aid that helps someone with physical impairments, such as limited wrist mobility, fish independently. Aimed at being more affordable than existing assistive fishing equipment on the market. Manufactured from mild steel, ash and ABS, using plasma cutting, wood turning and 3D printing.



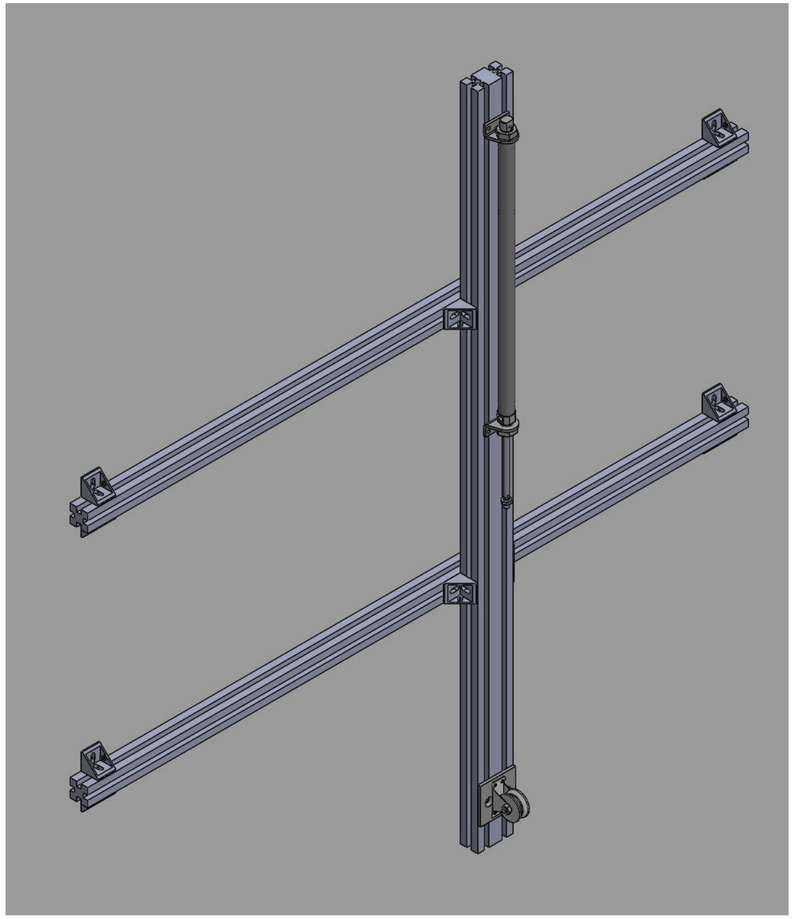


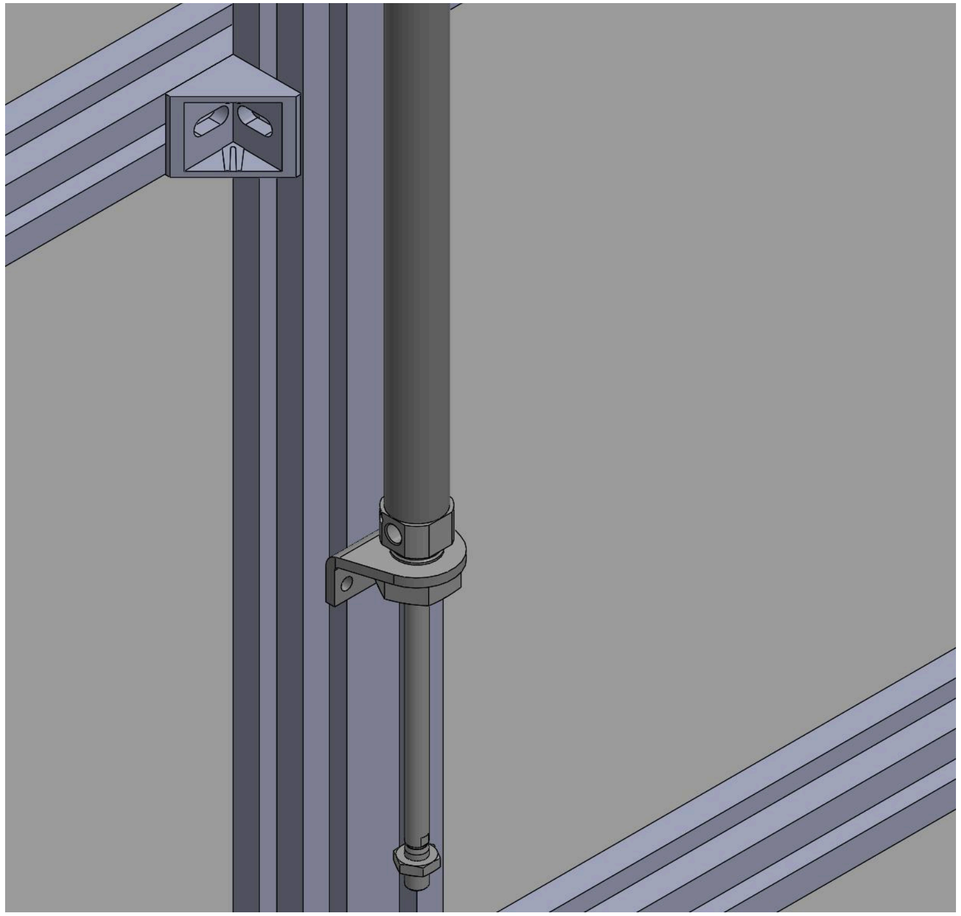


HIPER Test Rig

UROP, 2026

Designed a novel pneumatic catch and recovery system for the HIPER helmet rating scheme's test rig, as part of a UROP at Imperial's HEAD Lab. The design makes use of off the shelf components exclusively, yet still permits horizontal and vertical adjustment for testing with different anvils. I worked closely with researchers within the lab to establish a clear set of requirements for the design, ensuring that they were happy with the final product.



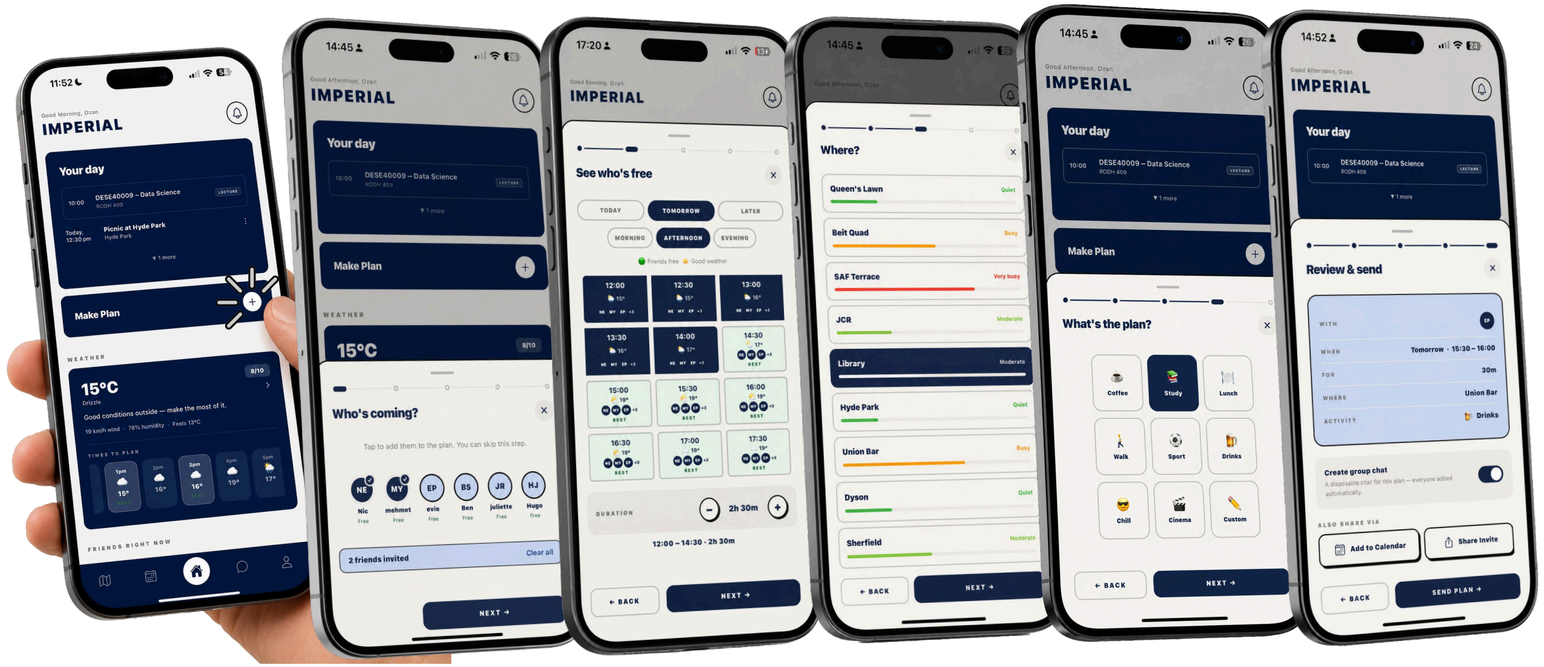


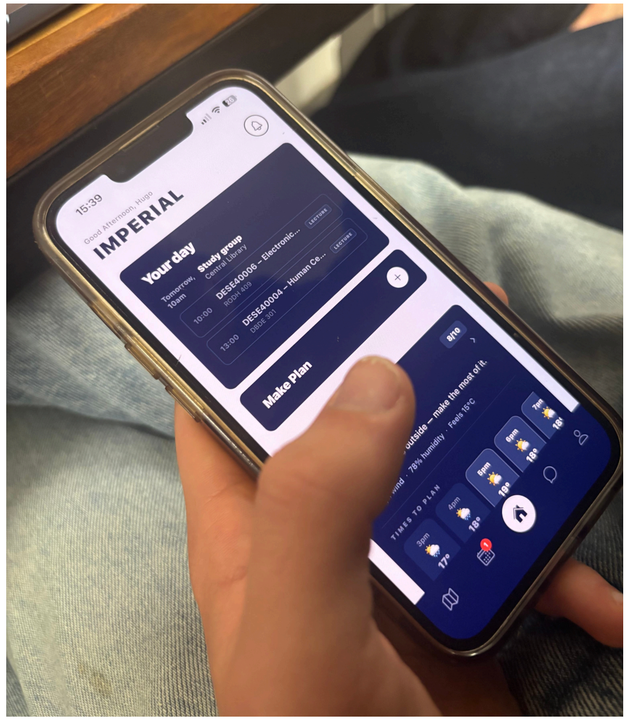


Cooped Up

Human-Centred Design, 2026

A social-planning app designed for my Human-Centred Design module, addressing how timetabling and weather act as barriers to student social life. The app lets users see who's free, check live weather and how busy campus spots are, then build and send a plan: picking a time and place and inviting friends in a few taps.









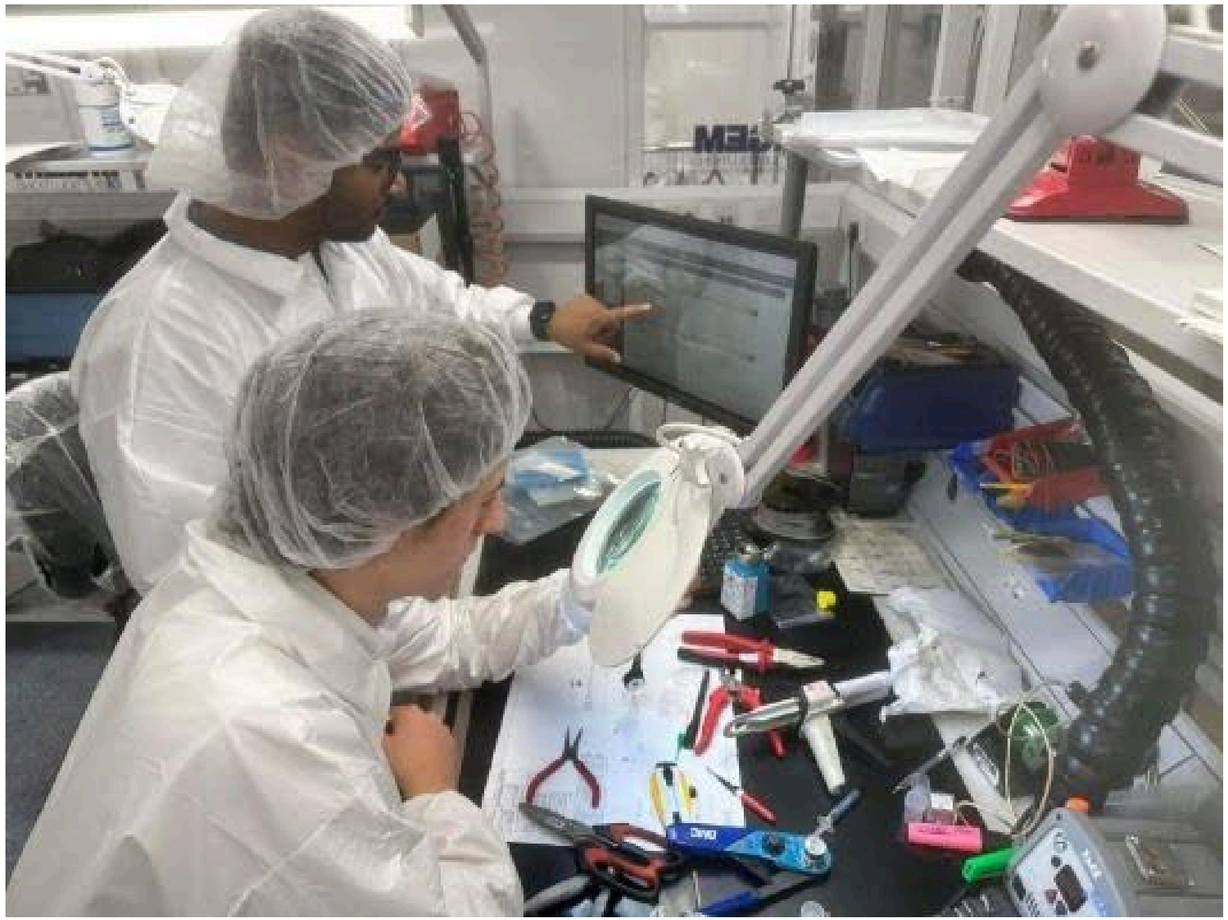
Research included weather diaries and interviews with students to understand when and why plans fall through.

Luka Hinton

Mercedes F1

GEM Cable Solutions, 2025

During my time at GEM Cable Solutions, I supported manufacture and quality control of cable assemblies for Mercedes F1, working to AS9100D aerospace-grade standards in a clean room environment.



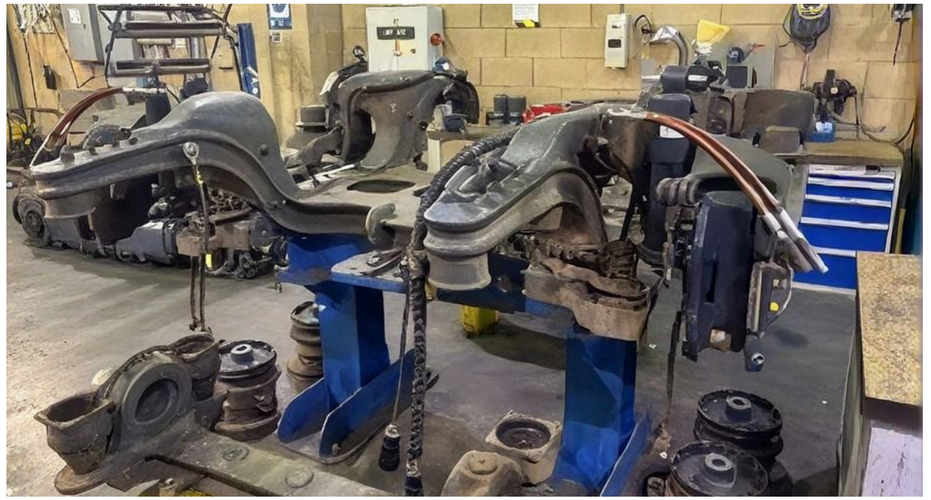


West Midlands Metro

GEM Cable Solutions, 2025

During my gap year at GEM Cable Solutions, I helped overhaul cable assemblies on the bogies of West Midlands Metro trams, working directly with client hardware pulled from service. This involved understanding the existing design, identifying the required components and sourcing alternatives as required.







Raleigh Scorpio Rebuild

Personal, 2026

Restoring a 1980s Raleigh Scorpio steel road bike bought for £30 at auction. Work so far has included a new chain and tyres, fixing a ball-bearing and freewheel fault and repairing faulty downtube shifting. Currently working toward a single-colour rattle-can respray with replica decals.



Fitness Dashboard

Personal, 2026

A personal training dashboard pulling in Hevy and Garmin data, with a weekly training planner and running-coach tools including an AI coaching assistant. Designed and refined through iterative prompting, directing the layout, features and data integrations rather than writing the code by hand.

